SWASTHYATWIN AI — TECHNICAL SYSTEM ARCHITECTURE

Multimodal Telemetry Ingestion • Causal BiDSleep ML • Mathematical Transparency • MongoDB Persistence

CLIENT LAYER (React 19 + TypeScript + Vite + TailwindCSS)

- Spatial Human Hologram Locator (SVG Vectors, 6 Subsystems: Keys 1-6)
- Multi-Signal Timeline with Provenance Badges
- Universal 'How is this calculated?' Explainability Modals
- What-If Counterfactual Simulator Workspace
- Sleep Intelligence & Dual-Track EEG vs PPG Hypnogram
- Real-Time MongoDB Health & Latency Probe

REST / JSON over HTTPS (Pydantic v2 Contracts)

FASTAPI APPLICATION GATEWAY & SECURITY

- CORS Whitelisting & Constant-Time HMAC Bearer Token / API-Key Verification
- Multi-Tenant Isolation: Mandatory {'patient_id': id} Query Scoping on All Operations
- AI Plausibility Guard: Typo Rejection (e.g., HR=700 bpm or SBP < DBP flagged 409 Conflict)
- Public Path Sanitization: Local Server Directory Scrubbing & Credential Masking in Logs

ANALYTICS & COMPUTATION ENGINE

- Robust Baseline Corridor: nMAD = 1.4826 * MAD
- Target Self-Exclusion & >=14 Obs Minimum Gate
- Personal Z-Score Anomaly Detector (Floor >= 1.0 bpm)
- Explainable Vitality Index Deduction Model (0-100)
- What-If Exponential Saturation Recovery Kinetics

MACHINE LEARNING INFERENCE ENGINE

- PhysioNet BiDSleep v1.0.1 (31 Subjects, 15.7k Epochs)
- 12 Causal Temporal Features (0% Future Data Leakage)
- Class-Weighted Random Forest (150 trees, depth 12)
- Held-Out Evaluation: Macro F1 28.22% (+18.69% lift)
- Sensor Dropout Guard: valid_count < 2.0 -> Unknown

DATA PERSISTENCE LAYER (MongoDB 7.0 + In-Memory Fallback)

Collections: db.patients (Demographics) | db.observations (Canonical Units) | db.health_assessments (Decay)
db.simulations (Isolated What-If Scenarios) | db.alerts (Deduplicated Flags) | db.audit_events (Scrubbed)